

MANISH KUMAR YADAV

manishky.official@gmail.com ♦ +9779844379957 ♦ [GitHub](#) ♦ [LinkedIn](#)

EDUCATION

Pulchowk Campus, IOE, Tribhuvan University

Bachelor's Degree in Electronics, Communication and Information Engineering
Grade: 79.5%

Lalitpur, Kathmandu

Apr 2021 - Apr 2025

PROFESSIONAL EXPERIENCE

Telemed Nepal



Full Stack Developer: TypeScript, React, PostgreSQL, Supabase

Feb 2026 - Present

- Built a three-role app (patient, doctor, admin) with Google OAuth and email auth backed by Supabase RLS
- Implemented a multi-state consultation workflow with Supabase Realtime subscriptions for live status updates across all dashboards
- Handled file uploads for prescriptions and payment receipts
- Served 1200+ visits and 28 consultations in first 2 months

Himalaya College of Engineering, Tribhuvan University

Kathmandu

Assistant Lecturer

May 2025 - Apr 2026

- Taught Microprocessor, Data Communication, Digital Logic and Embedded Systems

HexCode Nepal

Kathmandu

ML Engineer Intern

Feb 2023 - Aug 2023

- Implemented supervised, unsupervised, and reinforcement learning algorithms in Python
- Developed spam filter algorithm for text messages using binary classification
- Developed Kathmandu land price prediction algorithm using linear regression

OPENSOURCE CONTRIBUTION

ns-3-dev: A discrete event network simulator (540 stars)



Contribution: (C++) Fixed outdated header path references in ns-3's documentation

Apr 2025

PROJECTS

shellsense: a CLI tool to explain Linux shell commands in plain English



Python

Mar 2026

- Implemented a command parser that segments flags, subcommands, and arguments
- Resolved explanation through multiple layers: knowledge base, cache, man pages and tldr
- Rendered structured terminal output using the Rich library

job-aggregator: a job listings scraper



Python

Apr 2026

- Built a Python automation tool that scrapes job listings from multiple sources, removes duplicates, scores the listings against resume keywords, and emails a daily digest using GitHub Actions

p2p-transfer: browser based app to send files directly between two devices



JavaScript, Node.js, WebRTC

Dec 2025

- Built WebRTC p2p file transfer with a Node.js WebSocket signaling server
- Streamed files in chunks with backpressure control to handle large files
- Added SHA-256 password-protected rooms, QR code sharing, and real-time transfer speed display

RISC single cycle and pipelined processor



Verilog, GTKWave

Sep 2025

- Implemented program counter, instruction memory, register file, data memory, arithmetic logic unit, and control unit

TECHNICAL SKILLS

- **Programming:** Python, C++, C, JavaScript, Assembly, Shell, SQL, Verilog, MATLAB
- **Developer Tools:** VS Code, Git/GitHub, Docker
- **Framework & Libraries:** React, Node.js, WebRTC, Supabase
- **Technical Skills:** Linux

RESEARCH EXPERIENCE

MDPI JCP

 [Web](#)

10 citations, 6800 reads

Mar 2025

- Machine Learning-Based Attack Detection and Mitigation with Multi-Controller Placement Optimization over SDN Environment

IEEE Access

 [Web](#)

1 citation, 1133 reads

Mar 2025

- Game Theoretic Approach to QoS Oriented Machine Learning Model Development towards 5G Network Migration Planning.

CERTIFICATIONS

- [Coursera: Machine Learning Specialization \(Stanford | Online\)](#)
- [Coursera: Python for Everybody Specialization \(University of Michigan\)](#)